

KUMAR HARSH

Data Scientist

📍 Ranchi, Jharkhand | ✉️ kh949118@gmail.com | ☎️ +91-9279157296
in [LinkedIn](#) | [GitHub](#) | [HackerRank](#) | [LeetCode](#) | [Codolio](#)

Summary

Data Scientist & AI/ML Engineer with an M.Tech in Data Science and a B.Tech in Mechanical Engineering, combining strong engineering fundamentals with expertise in **machine learning, deep learning, data analysis, and computer vision**. Proficient in **Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, and Transformers**, with hands-on experience developing **end-to-end ML and AI solutions** for prediction, classification, computer vision, and intelligent automation. **GATE 2023 qualified** in Mechanical Engineering with **All India Rank 8240** and possess **1 year of industry experience** at **Larsen & Toubro (L&T) Hydrocarbon Engineering Limited**, contributing to offshore structural projects for IOCL and ARAMCO. Experienced in transforming real-world problems into **data-driven solutions** through statistical analysis, predictive modeling, and AI-based systems, with a strong focus on **model performance, scalability, and practical business impact**.

Technical Skills

Generative AI & Agents

- Generative AI & LLMs
- Multi-Agent AI Systems
- RAG & Agentic AI
- Prompt Engineering
- AI Agents & AI Workflows
- Claude, Gemini, ChatGPT
- Cursor & AI-Assisted Development

AI/ML & Data Science

- Machine Learning & Deep Learning
- Scikit-learn, XGBoost
- TensorFlow, PyTorch
- Pandas, NumPy, SciPy
- NLP, Transformers
- EDA, Feature Engineering, Model Evaluation
- Statistical Analysis & Data Visualization

Practical AI Development

- Python, FastAPI, Streamlit
- REST APIs & AI APIs
- Computer Vision Applications
- OCR & Object Detection
- Roboflow, YOLO, OpenCV, Easy-OCR
- MongoDB, MySQL, Git, Docker
- MERN Stack & Azure

Professional Experience

Larsen & Toubro Hydrocarbon Engineering

[Offer Letter]

Surat, Gujarat

Data Analyst – Offshore Structural Engineering

Jul 2023 – Jun 2024

- Analyzed structural integrity of **jackets and modules** for IOCL, ARAMCO & LINDE projects using **MS Excel** for joint connection validation and data quality checks
- Verified **joint entry accuracy and material specifications** in Staad Pro under seismic loading/unloading conditions, reducing design errors by **15%**
- Designed structural components in **AutoCAD & Tekla**, optimizing load capacity and durability for offshore platforms
- Supported multi-million dollar projects ensuring compliance with international offshore engineering standards

Internships & Vocational Training

Tata Steel

Jamshedpur, Jharkhand

Intern – Study of G Blast Furnace Cooling System

Dec 2021 – Feb 2022

- Conducted an in-depth study of the **cooling system** of the 'G' Blast Furnace, a critical component for maintaining operational efficiency and extending campaign life.
- Analyzed cooling plate performance, water circulation dynamics, and heat flux parameters to understand system effectiveness and identify potential failure modes.
- Prepared a detailed technical report correlating cooling system health with furnace operational data, contributing to a better understanding of maintenance requirements.

[Certificate]

SAIL (Steel Authority of India) – R&D Centre

Ranchi, Jharkhand

Summer Training – Rolling Process & Material Testing

July 2022

- Gained hands-on exposure to the industrial **rolling process** for steel production, understanding its impact on material microstructure and mechanical properties.
- Participated in **fatigue testing** and **residual stress measurement** experiments on steel samples, learning critical material characterization techniques.
- Correlated processing parameters with material performance, enhancing knowledge of steel metallurgy.

[Certificate]

Heavy Engineering Corporation (HEC)
Vocational Trainee – Manufacturing Processes

Ranchi, Jharkhand
Nov – Dec 2021

- Completed a comprehensive training program covering core manufacturing processes including large-scale **forging, casting, heat treatment, and machining** operations for heavy engineering equipment.
- Observed various **heat treatment** processes (annealing, normalizing, quenching, tempering) and their effect on material properties.
- Gained practical knowledge of **machining** operations on heavy-duty lathes and milling machines.

[\[Certificate\]](#)

IIRS, ISRO Outreach Programme
Workshop on Remote Sensing Technology

Dehradun (Online)
Oct – Nov 2021

- Certified in a foundational course on Remote Sensing and its diverse applications for geological mapping, environmental monitoring, and infrastructure planning.

[\[Certificate\]](#)

SkYY Rider Institutions

(Online)

Workshop on Basics Terminology of Suspension System

Apr 2022

- Participated in a two-day virtual workshop covering fundamental concepts of **automotive suspension systems**, including components, geometry, and design considerations.

CERTIFICATE NO: SRI-OT01-22440016529

[\[Certificate\]](#)

Featured Projects

Parking Lot Management System

Java, Spring Boot, Python, MySQL, JavaScript

- Engineered scalable system using **HashMap** for O(1) vehicle lookups and **Heap** for optimal slot allocation
- Implemented **Dynamic Programming** algorithm to calculate minimum parking fees, optimizing cost for customers
- Designed real-time dashboard with **live slot tracking** and transaction history using JavaScript
- Architected fault-tolerant backend with **automatic failover** to local storage when database is offline
- [\[Live Demo\]](#)

ML Studio Pro - Enterprise ML Platform

Python, Flask, scikit-learn, JavaScript

- **Upload & Analyze:** Cleans CSV/Excel datasets (up to 10GB) in **<2 mins**—auto-handles missing values & duplicates
- **Train & Compare:** Runs **8+ algorithms** (XGBoost, Neural Networks) on your data with real-time accuracy (85-94%), precision, recall & F1 scores
- **Predict & Deploy:** Single/batch predictions via simulated API with model version tracking (Production/Staging)
- **Visualize & Report:** Interactive charts + **6-page PDF reports** with dataset overview, model performance & business insights
- [\[Live Demo\]](#)

FinanceTrack Pro

HTML, CSS, JavaScript, APIs

- Live dashboard to **track and trade** 20+ assets including **Bitcoin, Ethereum, Gold, Silver, Crude Oil, Indian stocks (Reliance, TCS, HDFC) and global stocks (Apple, Microsoft, NVIDIA)**
- Prices update automatically every **5 seconds** with **green/red indicators** showing profits/losses
- Click any card to **buy/sell**, build your portfolio, track P&L, and monitor **order history** — all saved in your browser
- [Try Live Demo](#)

Object Detector AI - 80+ Classes

Python, FastAPI, COCO Dataset, OpenCV

- Engineered real-time detector identifying **80+ COCO objects** at **30+ FPS** with 85% accuracy
- Integrated live camera processing with confidence scoring and bounding box visualization
- Optimized for web with **sub-100ms latency** and responsive UI
- [\[Live Demo\]](#)

MultiAI Chatbot Platform - 5 Specialized Assistants

Python, FastAPI, Gemini API, Vercel

- Engineered **5 domain-specific AI assistants** with context-aware switching and sub-2-second response times
- Implemented **real-time RAG pipeline** processing PDF/TXT uploads for intelligent document Q&A
- Designed **multi-turn conversation memory** maintaining context across 50+ message exchanges
- Optimized API calls achieving **40% cost reduction** through prompt caching and token management
- [\[Live Demo\]](#)

Additional Projects

- **Eco Analyzer** [Multi-Agent AI]
- **AI QuizMaster** [Gemini API]

- [Link](#) • **EmotionSense AI** [CNN/YOLOv8]
- [Link](#) • **Patient Data Platform** [Power BI]

- [CNN](#) | **YOLOv8**
- [Link](#)

- **Google Play Analytics** [PySpark]

- **Fraud Detection** [Scikit-Learn]
- **Stock Market Analyzer** [Pandas]
- **Diabetes Prediction** [React/Python]
- **Ecommerce Platform** [MERN]
- **Population Dashboard** [Tableau]

[Link](#)
[Link](#) | [Data](#)
[Link](#)
[Link](#)
[Link](#)

Education

M.Tech in Data Science

Indian Institute of Information Technology, Bhopal

CGPA: 8.38

Aug 2024 – Mar 2026

B.Tech in Mechanical Engineering

Birsa Institute of Technology Sindri, Dhanbad

CGPA: 8.55

Apr 2019 – May 2023

CBSE Class 12

Delhi Public School, Ranchi

84%

2019

CBSE Class 10

Notre Dame School, Gumla

CGPA: 10.0

2017

Certifications

Data Structures & Algorithms (UC San Diego) 95.66%

[\[Verify\]](#)

Google Data Analytics Certificate

[\[Verify\]](#)

Microsoft Azure Fundamentals (AZ-900)

[\[Verify\]](#)

Power BI Certification (Simplilearn) - ID: 9937403

[\[Verify\]](#)

Qualcomm AI Technical Foundation - ID: 6KvK6wIMrA

[\[Verify\]](#)

Data Analytics (Simplilearn) - ID: 9935675

[\[Verify\]](#)

EF SET English Certificate 71/100 (C2 Proficient)

[\[Verify\]](#)

Data Visualization & Analysis with R

[\[Verify\]](#)

Achievements

- **Silver Medalist:** NPTEL Solid Mechanics course (structural analysis proficiency)
- **City Rank 1:** National Science Olympiad (Class 10) and International Math Olympiad (Class 9 & 10)
- **Academic Excellence:** 1st Position consecutively from Class 6-10; District Topper (10.0 CGPA)
- **Extracurricular:** 1st Position in District-Level Badminton Championship

Extracurricular & Leadership

- **IBM Agentic AI Hackathon:** Developed innovative AI solutions in competitive environment
- **SELP by TEQUIP-III:** Selected for technical excellence and leadership program
- **ISRO Outreach:** Certified in Remote Sensing Technology by IIRS Dehradun
- **IIT-ISM Dhanbad Tech Fest (Concetto-19):** Contributed to Robotics & AI project showcase